



House Price Dataset — Exploratory Data Analysis Report

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INTRODUCTION

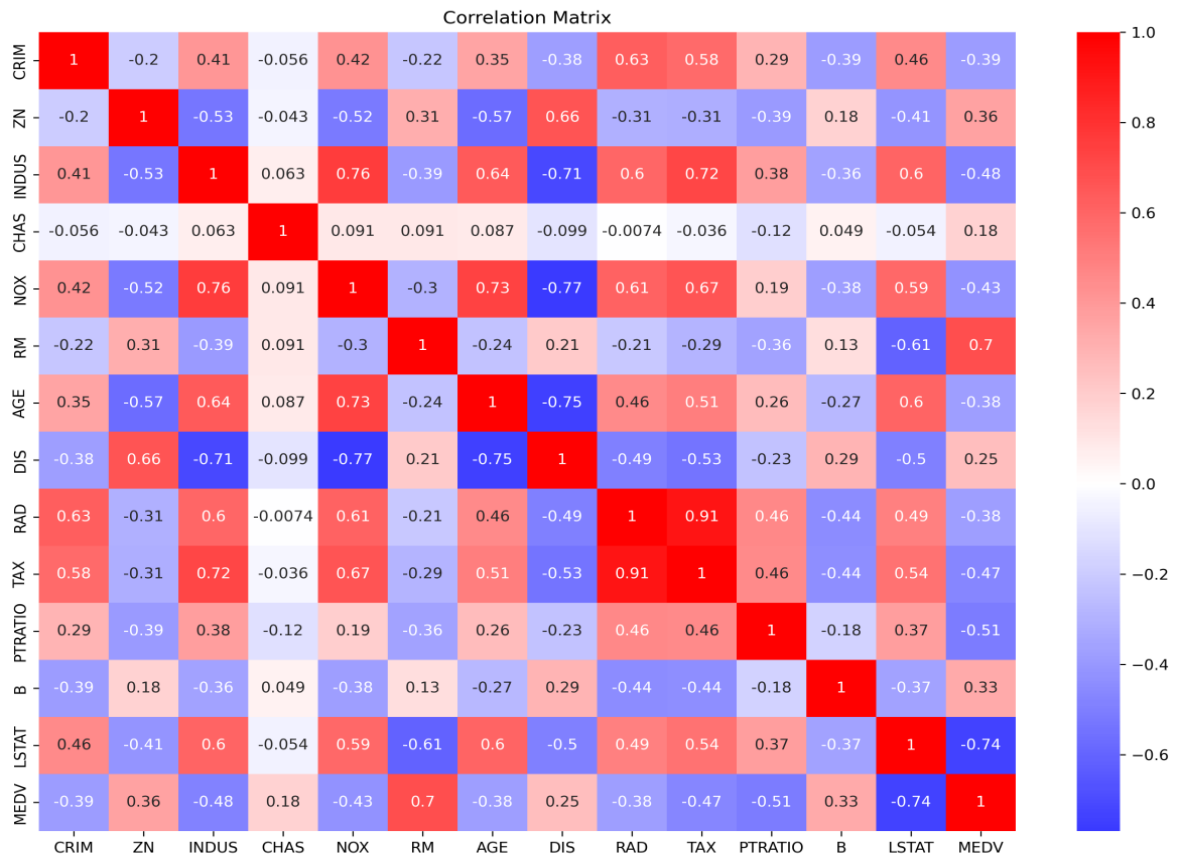
The dataset used in this analysis is the **House Price Prediction (Boston Housing)** dataset. It contains a set of neighborhood-level features such as crime rate, number of rooms, pollution levels, access to highways, and socio-economic indicators, alongside the target variable **MEDV**, which represents the median value of homes.

The main goal of working with this dataset is to understand **which factors have the strongest impact on house prices** and how these features interact. This analysis helps build a clear foundation for predictive modeling and for identifying the key drivers of housing value.

Summary of Key Findings

- Homes with **more rooms (RM)** tend to be significantly more expensive.
- Areas with **higher LSTAT** (lower socio-economic status) show sharp declines in home values.
- Proximity to the Charles River (CHAS)** is associated with noticeably higher prices.
- MEDV has a **price ceiling at 50**, which indicates capped values in the dataset.
- Strong correlations such as **RAD-TAX** and **NOX-INDUS** highlight multicollinearity that must be handled during modeling.

Correlation Matrix Insights



Key Insights:

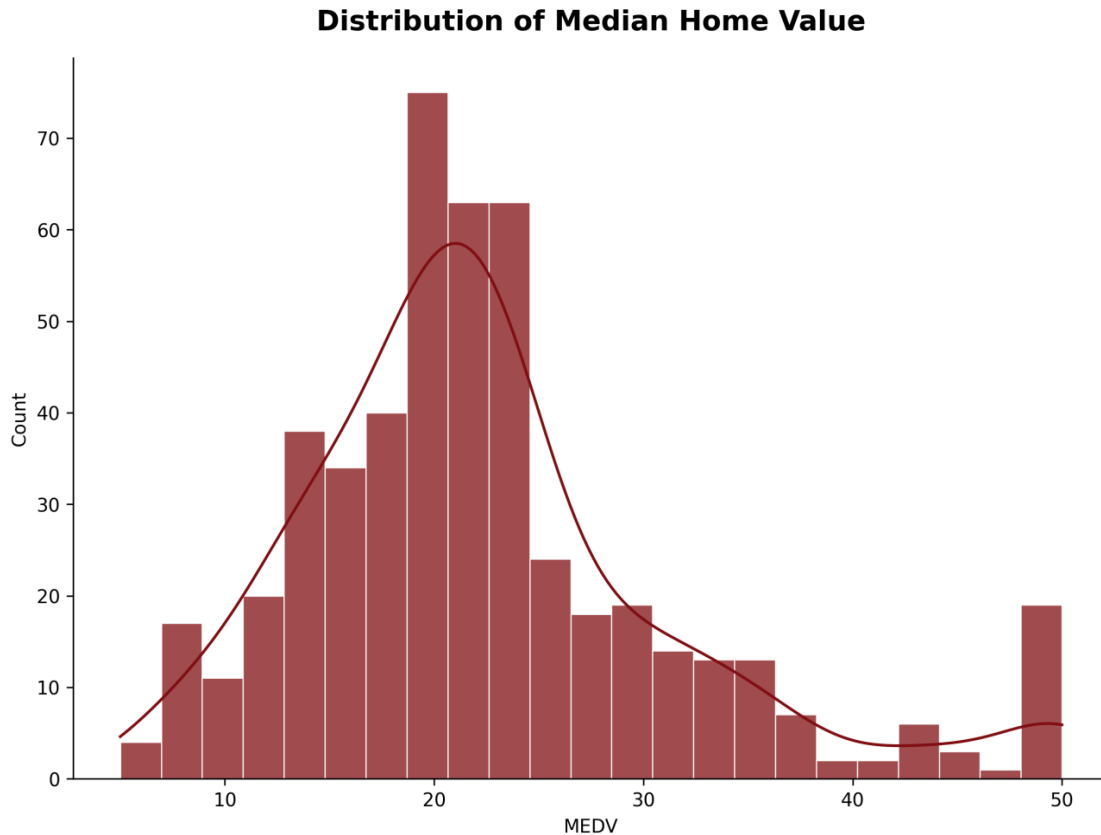
- There is a strong positive correlation between **RM** (number of rooms) and **MEDV** (house price)
→ Houses with more rooms tend to be more expensive.
- **LSTAT** shows a strong negative relationship with **MEDV**
→ Areas with a higher percentage of lower-status population tend to have lower house prices.
- **NOX** (pollution level) is negatively correlated with **MEDV**
→ Higher pollution is associated with lower property values.
- **CRIM** (crime rate) has a negative but moderate correlation with **MEDV**
→ Higher crime may reduce house prices, but the effect is not very strong.
- There is a very strong positive correlation between **RAD** and **TAX**
→ Areas closer to highways tend to have higher property taxes.

- **PTRATIO** has a negative relationship with **MEDV**
→ Higher student-to-teacher ratios are linked to lower house prices.
- **DIS** shows a weak positive relationship with **MEDV**
→ Distance from employment centers has a slight and unclear effect on prices.

Modeling Implication:

Prioritize RM and LSTAT as primary predictors. Handle multicollinearity (RAD-TAX, NOX-INDUS) via regularization or feature selection.

MEDV Distribution Insights



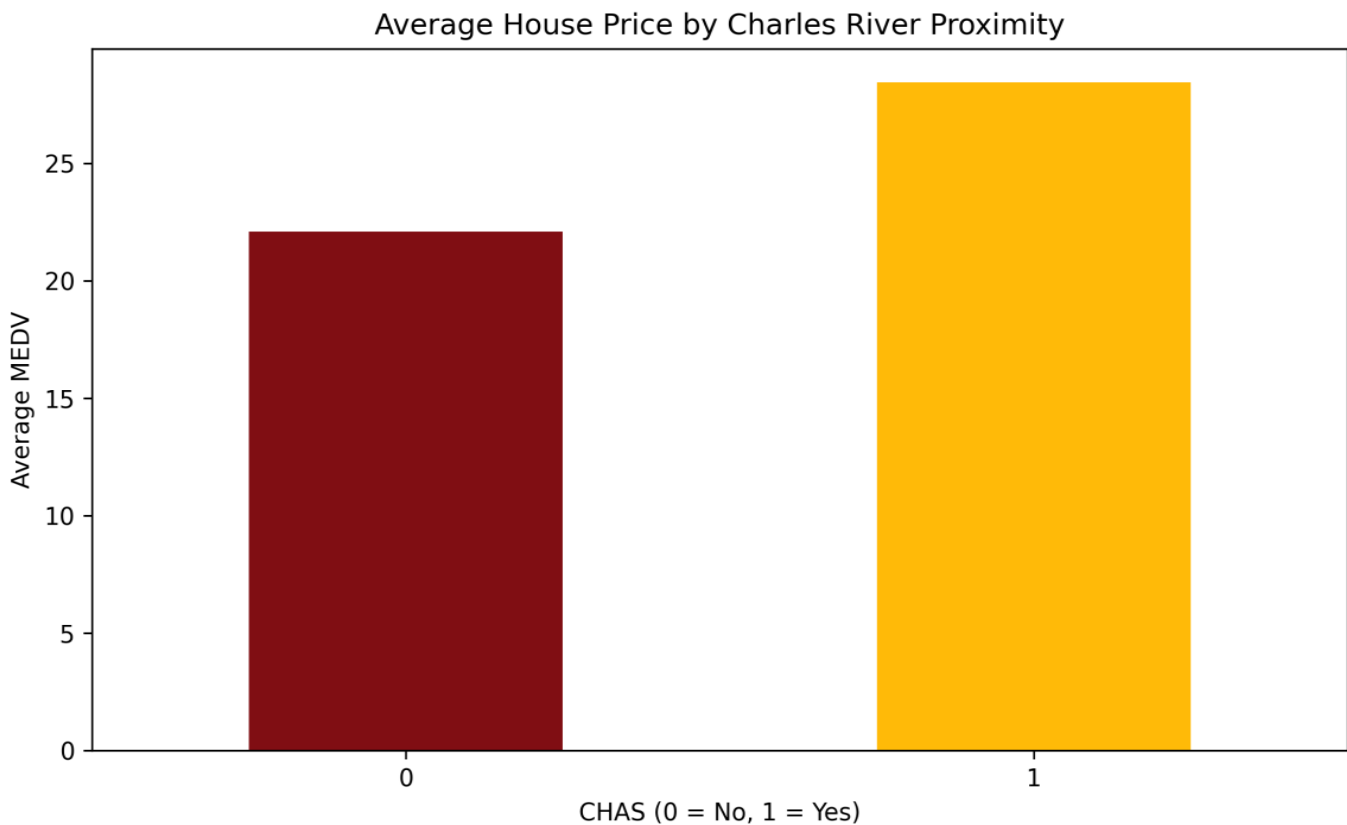
Key Insights:

- Most home values fall between \$15k and \$25k, indicating a predominantly mid-market housing environment.
- A hard cap near \$50k suggests price ceiling / censored values, especially for premium properties.
- Distribution is right-skewed, meaning high-value homes are underrepresented or limited.
- The market shows strong concentration in the middle range with few true high-end homes.

Modeling Implication:

The non-normal distribution and ceiling effect may bias linear model assumptions; consider transformations or models robust to censoring.

CHAS vs. MEDV Insights



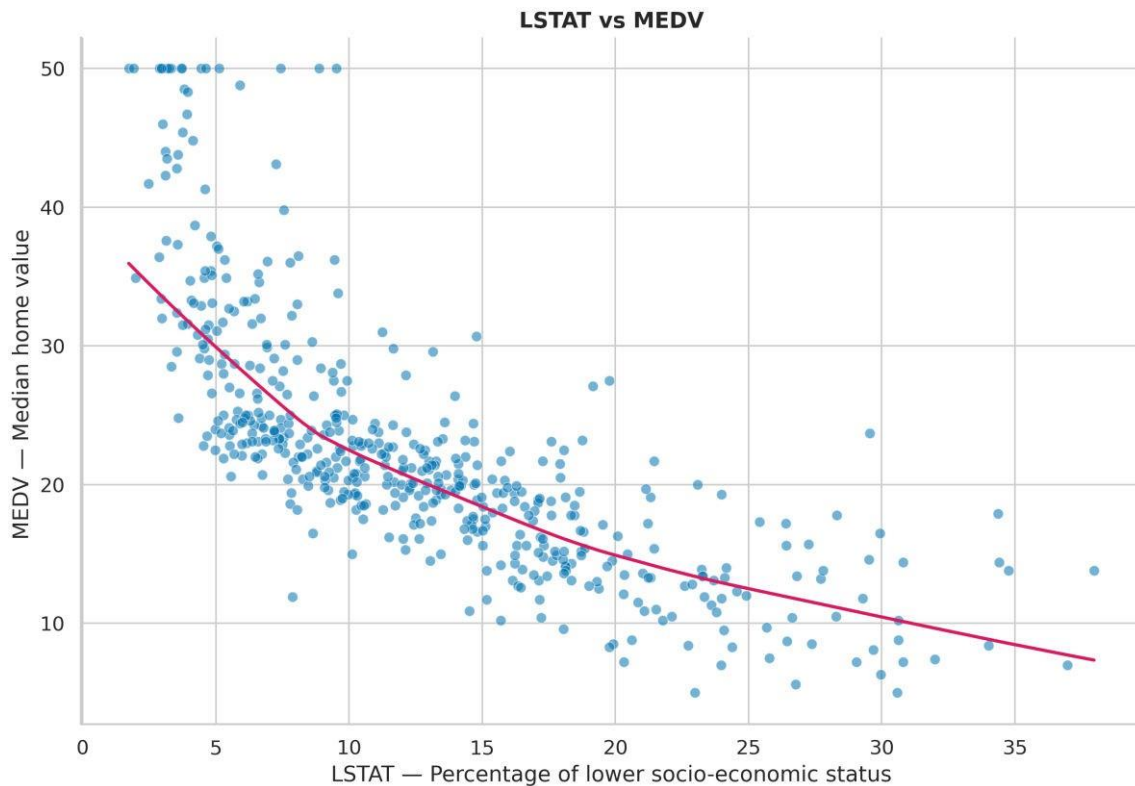
Key Insights:

- Houses near the Charles River (CHAS = 1) have a higher average home value than houses located farther away (CHAS = 0).
- The chart suggests that river proximity is associated with increased property prices in the dataset.
- This may reflect the added value of environmental attractiveness and premium location benefits.
- Although the difference is clear in the averages, CHAS is a binary feature and should be interpreted alongside other important housing variables.

Modeling Implication:

Treat CHAS as a meaningful categorical feature, since proximity to the Charles River appears to have a noticeable influence on average house prices.

LSTAT vs. MEDV Insights



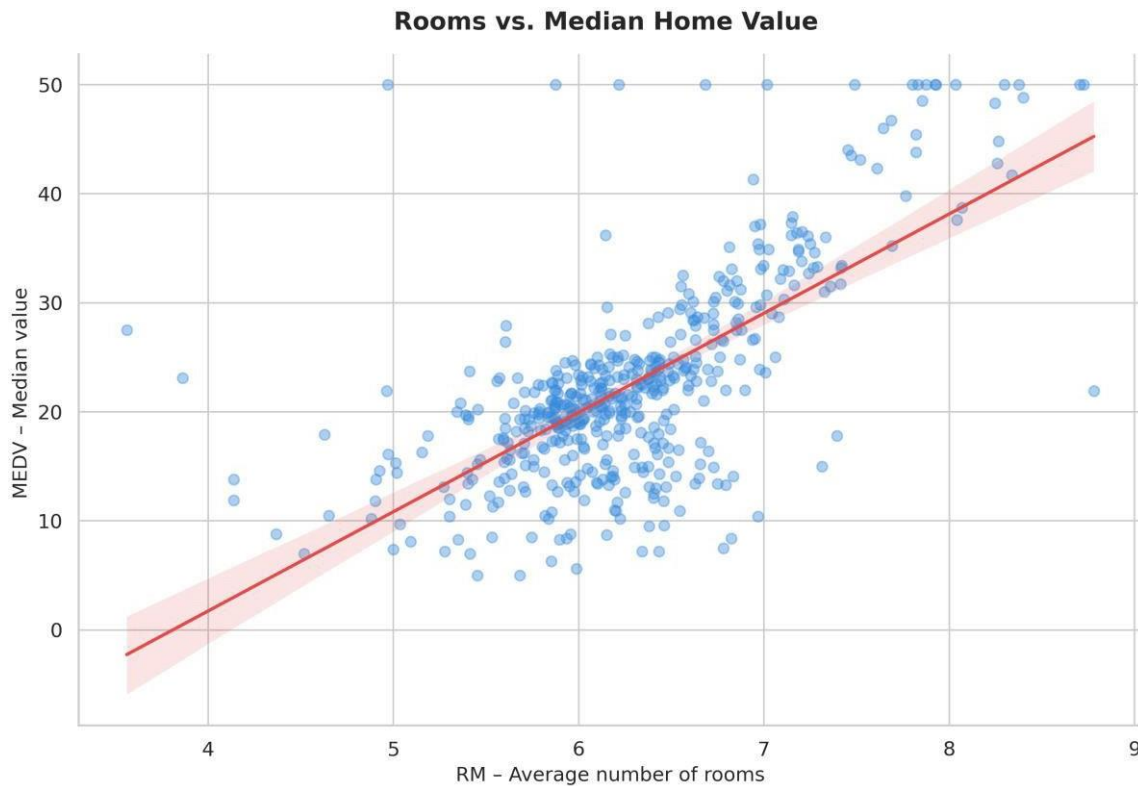
Key Insights:

- The relationship is strongly non-linear — price drops faster at higher LSTAT levels.
- Low LSTAT (3–8%) areas consistently contain the highest-valued homes.
- Once LSTAT exceeds ~20%, property values fall sharply and remain low regardless of other features.
- LSTAT is one of the most powerful socio-economic predictors of housing value.

Modeling Implication:

Spearman correlation tends to be stronger than Pearson here; allow curvature in models (e.g., polynomial terms or tree-based models).

RM vs. MEDV Insights



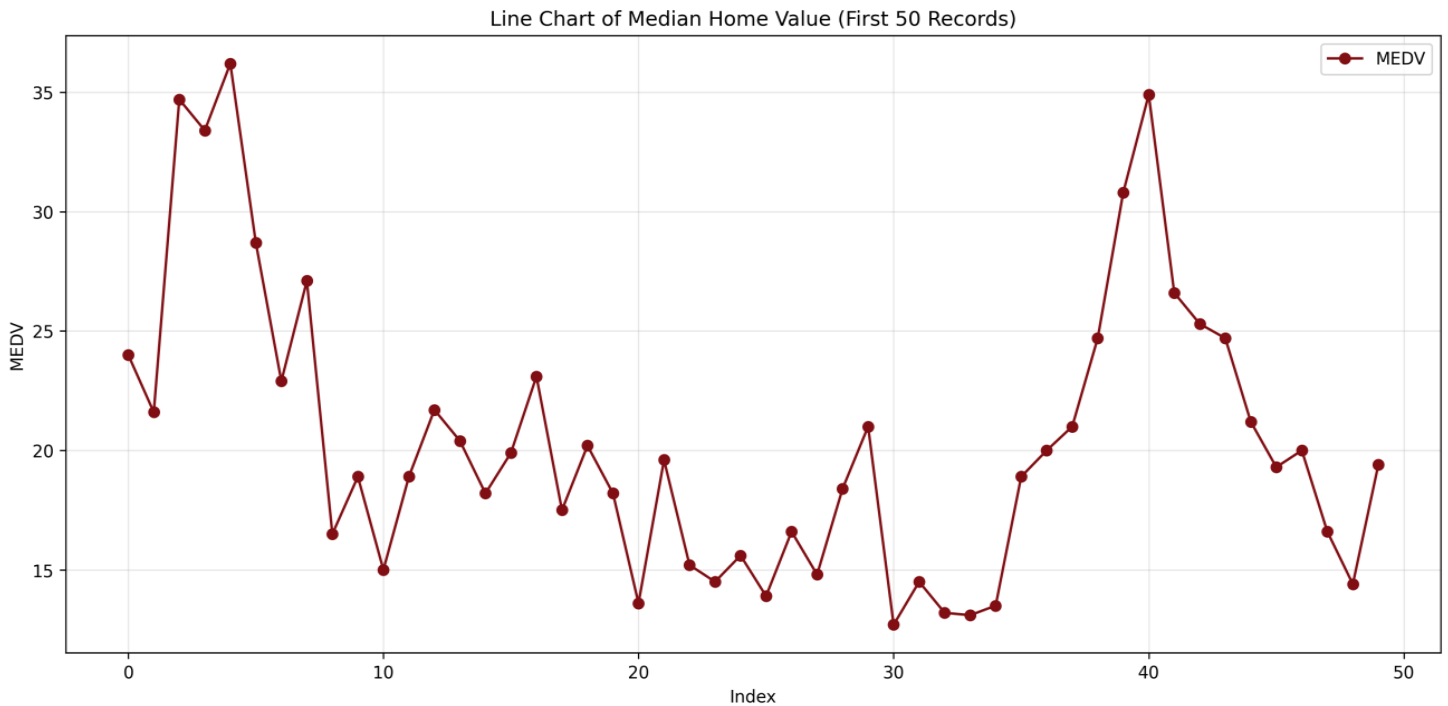
Key Insights:

- Strong linear positive relationship — more rooms directly increase price.
- Homes with 7+ rooms almost always fall in the high-value range.
- Homes with 4–5 rooms cluster at the low-to-mid value range.
- RM is a top predictor for modeling, showing clean, consistent upward trends.

Modeling Implication:

RM is ideal for linear regression and feature engineering (interaction or nonlinearity with LSTAT)

Median Home Value Across the First 50 Records



Key Insights:

- The line chart shows noticeable fluctuations in median home value (MEDV) across the first 50 records.
- House prices do not follow a steady upward or downward trend, indicating variability between observations.
- Several visible peaks and drops suggest that home values are influenced by multiple underlying factors rather than a simple sequential pattern.
- This chart is useful for visualizing short-range variation, but since the dataset is not time-based, the line should be interpreted as changes across records rather than a true time trend.

Modeling Implication:

The variation in MEDV across observations suggests that house prices are driven by a combination of features. Therefore, predictive modeling should rely on multiple explanatory variables rather than record order alone.